Ericsson PE Role Standard Operating Procedure for QoS Congestion Event (ALARM_7296) on JAY-PE-09 at Jayanagar Metro Site

Node: JAY-PE-09 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_7296

Description

The QoS Congestion Event (Microbursts, interface tail drops, QoS queue starvation) failure domain on the Ericsson PE role in the JAY-PE-09 node is attributed to the inherent characteristics of the Enhanced Packet Transport Protocol (EPT) employed in the 5500 series routers. EPT enables scalable packet transport while ensuring low-latency and high-throughput performance. However, this comes at the cost of increased complexity in the control plane, with intricate interactions between multiple protocols and processes. In the event of a microburst, the PE node's interface may enter a tail-drop state, causing QoS queue starvation. This can lead to a cascade of packet drops, ultimately resulting in the ALARM_7296 warning. The root cause of this failure domain lies in the delicate balance between packet forwarding rates and buffer sizing, which can be disrupted by transient network events or misconfigured QoS policies.

Pre-requisite Checks

- Dispatch a team of skilled engineers to the Jayanagar Metro Site for physical presence and assistance.

- Verify the status of redundant links and ensure they are operational and synchronized.

- Confirm that the Ericsson 5500 series router is running the latest software release and has the most recent configuration files applied.

Potential Root Causes

Cause	Probability
Misconfigured QoS policies leading to inadequate buffer sizing	High
Transient network events causing packet forwarding rate mismatch	Medium
Inadequate interface configuration resulting in tail-drop state	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces gigabitethernet 0/0/0 | include QoS queue,
Expected Output: Queue depth: 1000, Tail drop: 0, QoS policy: ,
Description: This command displays the current queue depth and tail
drop statistics for the specified interface. The expected output
indicates that the queue is not experiencing tail drops, and the QoS
policy is correctly configured.
```

```
Command: show qos policy , Expected Output: Policy name: , Buffers:
1000, Weight: 50, Description: This command displays the QoS policy
configuration, including the buffer size and weight. The expected
output indicates that the buffer size is set to 1000, which is
adequate for the current traffic load.
```

Resolution Steps

Step 1: Step: Adjust QoS policy to increase buffer size, Command: config qos policy buffers 1500, Description: This step involves modifying the QoS policy to increase the buffer size to 1500. This will provide additional capacity to handle transient network events and prevent tail drops.

Step 2: Step: Verify interface configuration for tail-drop prevention, Command: show interfaces gigabitethernet 0/0/0 | include tail-drop, Description: This step involves verifying that the interface is not configured to enter a tail-drop state under normal conditions. If the interface is configured to tail-drop, adjust the configuration to prevent this behavior.

Post-Check Verification

```
Command: show interfaces gigabitethernet 0/0/0 | include QoS queue,
Expected Output: Queue depth: 1000, Tail drop: 0, QoS policy: ,
Description: This command verifies that the queue is not experiencing
tail drops and the QoS policy is correctly configured after the
resolution steps.

Command: show qos policy , Expected Output: Policy name: , Buffers:
1500, Weight: 50, Description: This command verifies that the QoS
policy has been updated with the correct buffer size.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: undo config qos policy buffers 1500, Description: This step involves restoring the original QoS policy configuration to prevent further issues.

- Step: Verify original interface configuration, Command: show interfaces gigabitethernet 0/0/0 | include tail-drop, Description: This step involves verifying that the interface is configured to its original state before the resolution steps.